

# Great Wall

Coercion-resistant authentication

Yuri S. Villas Boas · author, Bitcoin BIP-450

# This is not a foreign problem

344

documented physical attacks on Bitcoin holders

12

of them in Brazil

2

this year

Florianópolis, 2017. Jameson Lopp's public registry.

**Our answer has  
been:**

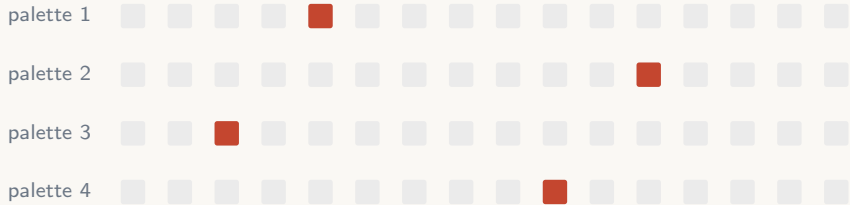
**“Don’t tell  
anyone.”**

# What actually works

You cannot be forced to hand  
over  
**something you cannot  
describe.**

Recognition, not recall. There is nothing to write down, and nothing to take.

# How it looks to a user



Four palettes. Sixteen faces each. Reshuffled every time.  
You pick the one you know.

Demo: [youtube.com/watch?v=9lJvw-8PZ0A](https://youtube.com/watch?v=9lJvw-8PZ0A)

# Two deployments, one protocol

## Custodial

$2^{16}$

The bank counts your guesses and freezes the account.

Sixteen bits is plenty.

## Self-custodial

$2^{160}$

*Nobody* counts guesses against your own encrypted seed.

Which is why it can't be faces.

# Where the money is

License it to banks

**\$80.8M**

serviceable market, 673 institutions

Brazil's central bank already imposes night-time Pix limits. It has conceded that **delay** is the countermeasure. Its own data shows the refund mechanism recovering ~9% of confirmed fraud.

Sell it to holders

**\$19.4M**

serviceable market

The self-custody marketplace. Smaller, and it is where the credibility comes from.

## The ask

**\$600K**

for 20% at \$3.0M post-money

17 months of runway — team, integration, first bank pilot



backup

# Why me

- ▶ Author of **BIP-450** — a Bitcoin standard, peer-reviewed by the same process
- ▶ **37-page paper** formalising the attack, with a public DOI
- ▶ Contributor to the attack registry the model is calibrated on
- ▶ great-wall-core is **public and in beta** — the encoder is real code

Solo founder. Pre-incorporation — I incorporate on term sheet, in whichever jurisdiction the round wants.

# “Doesn’t multisig solve this?”

No — it moves the problem.

A co-signer who can be reached is a second victim, not a second lock.

A co-signer who *cannot* be reached is a custodian, and you have given up self-custody to get there.

# “What about a duress PIN?”

That is the failure mode, not the fix.

A partial disclosure the attacker cannot verify gives them a reason to keep going. Graceful degradation — a virtue everywhere else in engineering — is, under coercion, an incentive to escalate.

Blockstream ships a duress PIN. That is the strongest evidence I have that the problem is real.

# Status, honestly

- ▶ Self-custodial encoder: **public beta**
- ▶ Custodial deployment: **designed, not built** — and it is the *smaller* engineering problem
- ▶ No entity yet; no revenue yet

The seed builds the custodial product and lands the first bank pilot.